

Lab 2

Network Commands for Testing and Troubleshooting

Objective

To study and use basic network commands for testing connectivity, configuration, and troubleshooting network-related problems in a computer system.

Requirements:

- Active Network connection
- Command Prompt

Theory

Network commands are essential tools provided by the operating system to monitor, test, configure, and troubleshoot computer networks. These commands help users understand how a system is connected to a network and how data is transmitted between devices. By using network commands, issues such as connection failure, incorrect IP configuration, routing problems, and DNS resolution errors can be easily identified and resolved.

Network Commands

1. ping

Checks whether a remote host is reachable and measures response time.

Syntax:

```
ping <IP address / domain name>
```

2. ipconfig

Displays IP address, subnet mask, and default gateway of the system.

Syntax:

```
ipconfig
```

3. tracert

Traces the path taken by packets from source to destination.

Syntax:

```
tracert <IP address / domain name>
```

4. netstat

Shows active network connections and listening ports.

Syntax:

```
netstat
```

5. nslookup

Resolves domain names into IP addresses.

Syntax:

```
nslookup <domain name>
```

6. arp -a

Displays the ARP table showing IP and MAC address mapping.

Syntax:

```
arp -a
```

7. telnet

Tests remote connectivity using the Telnet protocol.

Syntax:

```
telnet <IP address> <port number>
```

8. hostname

Displays the name of the current computer.

Syntax:

```
hostname
```

9. getmac

Shows the MAC address of network interfaces.

Syntax:

```
getmac
```

10. route print

Displays the routing table of the system.

Syntax:

```
route print
```

11. netsh

Configures and manages network settings.

Syntax:

```
netsh
```

12. systeminfo

Displays detailed system and network information.

Syntax:

```
Systeminfo
```

Results:

1. Ping

```
C:\Users\Acer>echo lab2_Aayuska
lab2_Aayuska

C:\Users\Acer>ping google.com

Pinging google.com [2404:6800:4007:805::200e] with 32 bytes of data:
Reply from 2404:6800:4007:805::200e: time=59ms
Reply from 2404:6800:4007:805::200e: time=58ms
Reply from 2404:6800:4007:805::200e: time=58ms
Reply from 2404:6800:4007:805::200e: time=58ms

Ping statistics for 2404:6800:4007:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 58ms, Maximum = 59ms, Average = 58ms
```

2. ipconfig

```
C:\Users\Acer>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2404:7c00:41:5634:6e27:b6d6:5b20:f1d3
    Temporary IPv6 Address. . . . . : 2404:7c00:41:5634:78b0:3fa0:7af3:ea5f
    Link-local IPv6 Address . . . . . : fe80::5579:8849:a7a9:6ea%5
    IPv4 Address. . . . . : 192.168.101.19
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::72a5:6aff:feb3:6194%5
                                192.168.101.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

3.Tracert

```
C:\Users\Acer>tracert google.com

Tracing route to google.com [2404:6800:4007:805::200e]
over a maximum of 30 hops:

  1    4 ms    3 ms    3 ms  2404:7c00:40::173:efbb
  2    6 ms    4 ms    5 ms  2404:7c00::3
  3    5 ms    5 ms    6 ms  2404:7c00:0:7::2
  4    7 ms    6 ms    6 ms  2404:7c00:0:6::2
  5   47 ms   47 ms   47 ms  2404:a800:3a00:2::279
  6   45 ms    *    45 ms  2404:a800::92
  7   64 ms   64 ms   64 ms  2404:6800:8201:3c0::1
  8   67 ms   64 ms   63 ms  2404:6800:8201:3c0::1
  9   47 ms   46 ms   47 ms  2001:4860:0:1::559e
 10  45 ms   46 ms   46 ms  2001:4860:0:1::1842
 11  91 ms    *    *    2001:4860:0:1::8829
 12  46 ms   46 ms   45 ms  2001:4860:0:1::564b
 13  58 ms   58 ms   58 ms  maa05s13-in-x0e.1e100.net [2404:6800:4007:805::200e]

Trace complete.
```

4.netstat

```
C:\Users\Acer>netstat

Active Connections

 Proto Local Address           Foreign Address         State
TCP    127.0.0.1:5141           LAPTOP-NK4TR378:50673  ESTABLISHED
TCP    127.0.0.1:49669         LAPTOP-NK4TR378:49670  ESTABLISHED
TCP    127.0.0.1:49670         LAPTOP-NK4TR378:49669  ESTABLISHED
TCP    127.0.0.1:49671         LAPTOP-NK4TR378:49672  ESTABLISHED
TCP    127.0.0.1:49672         LAPTOP-NK4TR378:49671  ESTABLISHED
TCP    127.0.0.1:49673         LAPTOP-NK4TR378:49674  ESTABLISHED
TCP    127.0.0.1:49674         LAPTOP-NK4TR378:49673  ESTABLISHED
TCP    127.0.0.1:50673         LAPTOP-NK4TR378:5141   ESTABLISHED
TCP    127.0.0.1:56055         LAPTOP-NK4TR378:56056  ESTABLISHED
TCP    127.0.0.1:56056         LAPTOP-NK4TR378:56055  ESTABLISHED
TCP    127.0.0.1:58995         LAPTOP-NK4TR378:62863  ESTABLISHED
TCP    127.0.0.1:58995         LAPTOP-NK4TR378:64916  ESTABLISHED
TCP    127.0.0.1:59057         LAPTOP-NK4TR378:59058  ESTABLISHED
TCP    127.0.0.1:59058         LAPTOP-NK4TR378:59057  ESTABLISHED
TCP    127.0.0.1:59059         LAPTOP-NK4TR378:59060  ESTABLISHED
TCP    127.0.0.1:59060         LAPTOP-NK4TR378:59059  ESTABLISHED
TCP    127.0.0.1:59735         LAPTOP-NK4TR378:59736  ESTABLISHED
TCP    127.0.0.1:59736         LAPTOP-NK4TR378:59735  ESTABLISHED
TCP    127.0.0.1:61851         LAPTOP-NK4TR378:61852  ESTABLISHED
TCP    127.0.0.1:61852         LAPTOP-NK4TR378:61851  ESTABLISHED
TCP    127.0.0.1:61853         LAPTOP-NK4TR378:61854  ESTABLISHED
TCP    127.0.0.1:61854         LAPTOP-NK4TR378:61853  ESTABLISHED
TCP    127.0.0.1:62863         LAPTOP-NK4TR378:58995  ESTABLISHED
TCP    127.0.0.1:64916         LAPTOP-NK4TR378:58995  ESTABLISHED
TCP    192.168.101.19:49677    spdns:domain           TIME_WAIT
TCP    192.168.101.19:50254    spdns:domain           TIME_WAIT
TCP    192.168.101.19:50634    spdns:domain           TIME_WAIT
TCP    192.168.101.19:51444    lb-140-82-113-26-iad:https ESTABLISHED
TCP    192.168.101.19:51484    20.42.65.94:https      ESTABLISHED
TCP    192.168.101.19:52973    spdns:domain           TIME_WAIT
TCP    192.168.101.19:53449    spdns:domain           TIME_WAIT
TCP    192.168.101.19:53626    spdns:domain           TIME_WAIT
TCP    192.168.101.19:53908    spdns:domain           TIME_WAIT
TCP    192.168.101.19:53986    spdns:domain           TIME_WAIT
```

5.nslookup

```
C:\Users\Acer>nslookup google.com
Server: UnKnown
Address: fe80::1

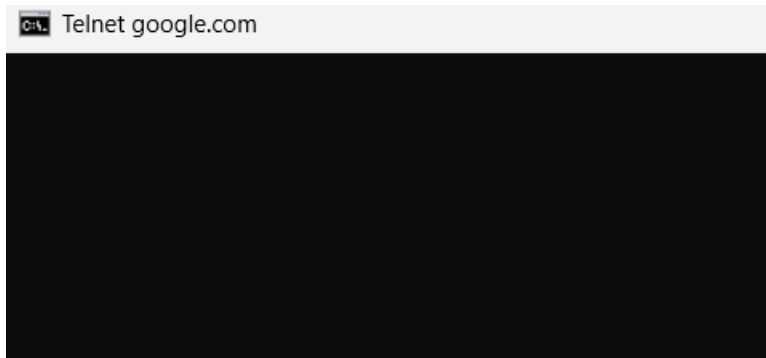
Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4007:805::200e
          142.251.220.110
```

6.arp -a

```
C:\Users\Acer>arp -a

Interface: 192.168.101.19 --- 0x5
Internet Address      Physical Address      Type
192.168.101.1         70-a5-6a-b3-61-94    dynamic
192.168.101.7         42-a3-15-61-1d-4b    dynamic
192.168.101.255       ff-ff-ff-ff-ff-ff    static
224.0.0.2             01-00-5e-00-00-02    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

7.telnet



8.hostname

```
C:\Users\Acer>hostname
LAPTOP-NK4TR378
```

9.getmac

```
C:\Users\Acer>getmac

Physical Address      Transport Name
=====
28-D0-43-78-C5-10     \Device\Tcpip_{2E9F375A-20F6-45B5-B57D-6D4600711CE3}
38-A7-46-0E-5F-B3     Media disconnected
28-D0-43-78-C5-11     Media disconnected
00-FF-68-14-09-45     Media disconnected
```

10.route print

```
C:\Users\Acer>route print
=====
Interface List
8...38 a7 46 0e 5f b3 .....Realtek PCIe GbE Family Controller
9...00 ff 68 14 09 45 .....TAP-Windows Adapter V9
18...2a d0 43 78 e5 30 .....Microsoft Wi-Fi Direct Virtual Adapter
7...2a d0 43 78 f5 20 .....Microsoft Wi-Fi Direct Virtual Adapter #2
5...28 d0 43 78 c5 10 .....RZ616 Wi-Fi 6E 160MHz
14...28 d0 43 78 c5 11 .....Bluetooth Device (Personal Area Network)
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.101.1    192.168.101.19   35
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.101.0              255.255.255.0    On-link          192.168.101.19   291
192.168.101.19            255.255.255.255  On-link          192.168.101.19   291
192.168.101.255            255.255.255.255  On-link          192.168.101.19   291
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          192.168.101.19   291
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          192.168.101.19   291
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
5 291 :::/0 fe80::72a5:6aff:feb3:6194
1 331 ::1/128 On-link
5 291 2404:7c00:41:5634::/64 On-link
5 291 2404:7c00:41:5634::/64 fe80::72a5:6aff:feb3:6194
5 291 2404:7c00:41:5634:6e27:b6d6:5b20:f1d3/128 On-link
5 291 2404:7c00:41:5634:78b0:3fa0:7af3:ea5f/128 On-link
5 291 fe80::/64 On-link
5 291 fe80::5579:8849:a7a9:6ea/128 On-link
1 331 ff00::/8 On-link
5 291 ff00::/8 On-link
=====
Persistent Routes:
None
```

11.Netsh

```
C:\Users\Acer>netsh
netsh>wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
-----
<None>

User profiles
-----
All User Profile      : Himalaya01
All User Profile      : HIMALAYA
All User Profile      : Computer Lab
```

1.

12.Systeminfo

```
C:\Users\Acer>systeminfo

Host Name:                               LADT00-M4MTR378
OS Name:                                 Microsoft Windows 11 Home Single Language
OS Version:                             10.0.22H2 N/A Build 22H2
OS Manufacturer:                        Microsoft Corporation
OS Configuration:                       Standalone Workstation
OS Build Type:                           Multiprocessor Free
Registered Owner:                        Acer
Registered Organization:                 N/A
Product ID:                              00382-42720-18788-A0DEM
Original Install Date:                   6/2/2025, 3:18:27 PM
System Boot Time:                         1/6/2026, 8:03:53 AM
System Manufacturer:                     Acer
System Model:                             Nitro ANV16-41
System Type:                             x64-based PC
Processor(s):                             1 Processor(s) Installed.
[01]: AMD64 Family 25 Model 117 Stepping 2 AuthenticAMD ~2499 Mhz
BIOS Version:                             Insyde Corp. V1.15, 10/29/2025
Windows Directory:                       C:\Windows
System Directory:                         C:\Windows\system32
Boot Device:                             \Device\HarddiskVolume1
System Locale:                             en-us;English (United States)
Input Locale:                             en-us;English (United States)
Time Zone:                               (UTC+05:45) Kathmandu
Total Physical Memory:                    16,384 MB
Available Physical Memory:                 6,956 MB
Virtual Memory: Max Size:                 24,892 MB
Virtual Memory: Available:                 6,463 MB
Virtual Memory: In Use:                    18,429 MB
Page File Location(s):                   C:\pagefile.sys
Domain:                                  WORKGROUP
Logon Server:                             \\LADT00-M4MTR378
Hotfix(s):                               3 Hotfix(s) Installed.
[01]: KB5066131
[02]: KB5072333
[03]: KB5071142
Network Card(s):                          4 NIC(s) Installed.
[01]: Realtek Wi-Fi 6E 160MHz
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 192.168.101.1
IP address(es)
[01]: 192.168.101.10
[02]: f4b0:5370:8400:a7a9:6ea
[03]: 2404:7c00:41:5634:78b0:3fa0:7af3:ea5f
[04]: 2404:7c00:41:5634:6e27:b6d6:5b20:f1d3
[02]: Realtek PCIe GbE Family Controller
Connection Name: Ethernet
Status: Media disconnected
[03]: Bluetooth Device (Personal Area Network)
Connection Name: Bluetooth Network Connection
Status: Media disconnected
[04]: TAP-Windows Adapter V9
Connection Name: McAfee VM
Status: Media disconnected
Virtualization-based security: Status: Running
Required Security Properties:
Base Virtualization Support
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMN Security Mitigations 1.0
Mode Based Execution Control
Services Configured:
Hypervisor enforced Code Integrity
Services Running:
Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```

Discussion

During the lab, various network commands were executed to understand their practical use in network testing and troubleshooting. Commands like `ping` and `tracert` helped verify connectivity and path analysis, while `ipconfig`, `netstat`, and `arp -a` provided configuration and connection details. These tools are essential for identifying network issues quickly and accurately.

Conclusion

The lab successfully demonstrated the use of network commands for testing and troubleshooting network problems. Understanding these commands helps in diagnosing connectivity issues, monitoring network performance, and maintaining efficient network operations.